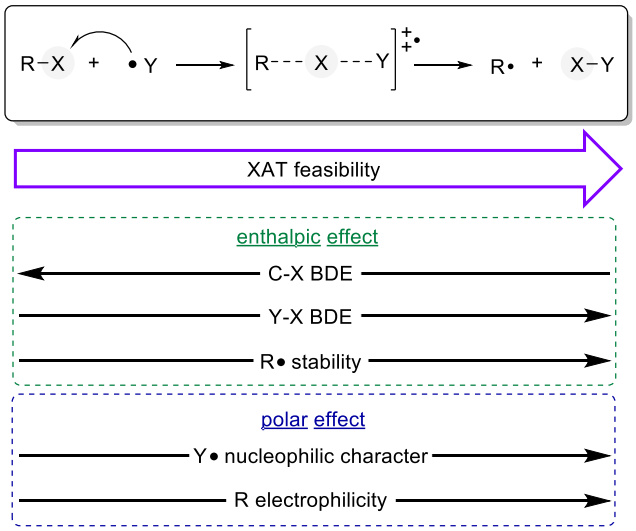
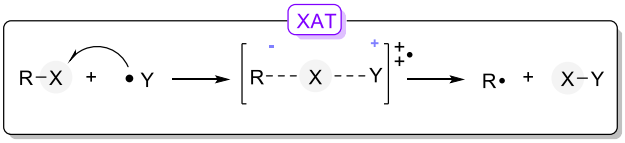
1. What are the key aspects that control reactivity in XAT reaction? A) redox potential B) bond-dissociation energy (BDE); C) polarizability of the C-X bond; D) solvent
   Answer: B and C are the correct answers. Bond-dissociation energy BDE (enthalpic effect) will tell you how easy it will be to cleave the bond and the polarizability of the C-X bond which is involved in the stabilizing charge-transfer character at the transition state will tell you how fast the halogen abstraction happens (kinetic effect).

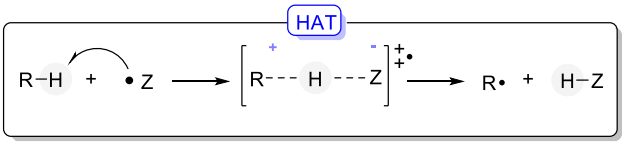


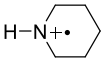
2. What is the main side-reaction in the XAT process? A) SET; B) HAT; C) β-scission; D) β-hydride elimination
   Answer: B is the correct answer. Indeed, hydrogen atom transfer (HAT) is the main side reaction which can prematurely reduce the new formed radical.

3. What type of radicals are used to initiate XAT processes? A) Nucleophilic; B) Electrophilic; C) Neutral
   Answer: The radicals used to initiate XAT reactions are nucleophilic (a partial negative charge of the carbon from which the halide is abstracted is built). Worth noting is that it is the opposite polar effect found in the HAT, in which the radical is electrophilic.



• Y : nucleophilic radical = • $SnBu_3$ • $Si(SiMe)_3$ • $GeBu_3$ • $Mn(CO)_5$

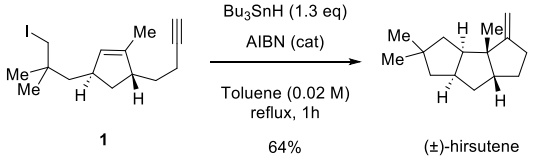


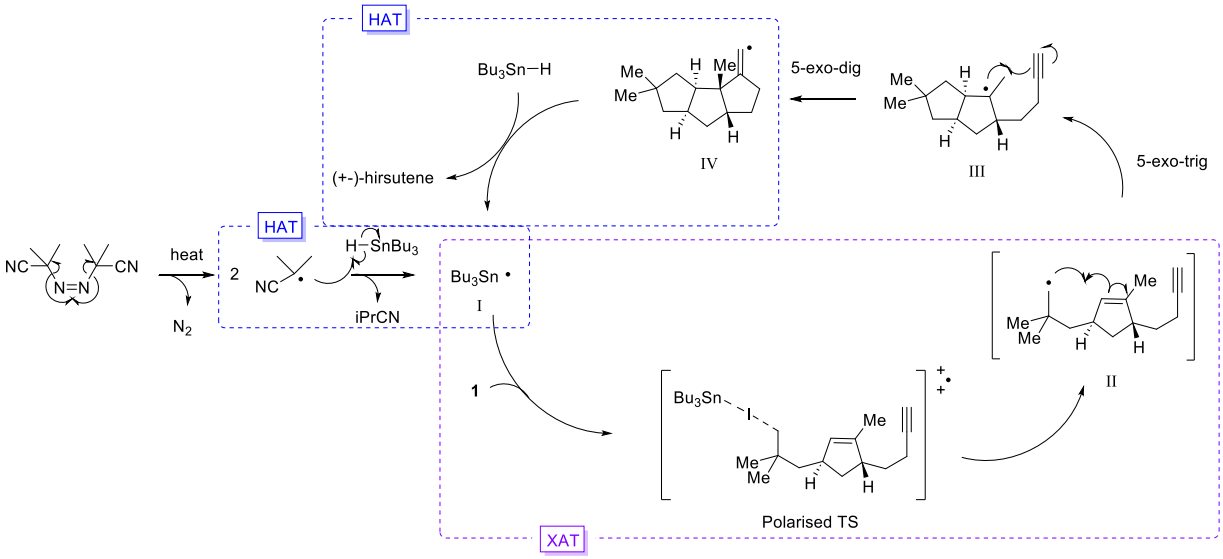
• Z : electrophilic radical = • OtBu • Ocumyl 

4. What are the two main methods to general radicals in XAT reaction?
   Answer: The two main methods to generate radicals in XAT reaction are thermal and photolytic. Worth noting that under thermal conditions, usually high dilution is required to avoid premature termination via HAT by tin hydride.

5. Dennis Curran and Donna Rakiewicz reported the total synthesis of Hirsutene (*J. Am. Chem. Soc.*, **1985**, *107*, 5, 1448-1449). Propose a mechanism of the following transformation.

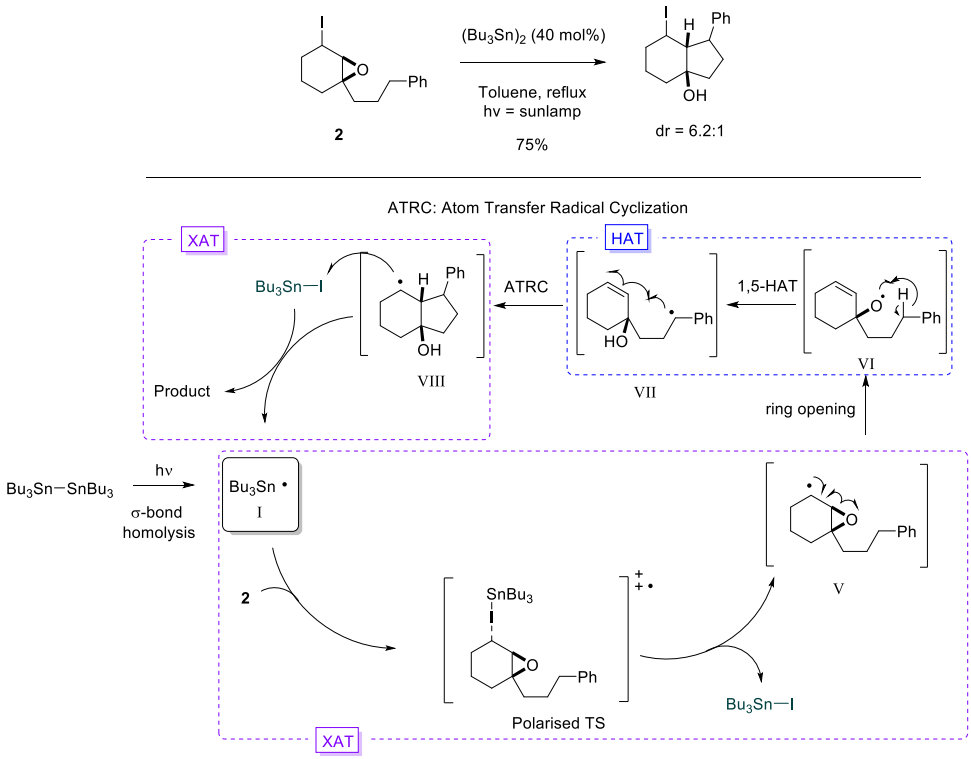
Answer: The tin radical *I* is generated from the initiation of AIBN via HAT process and reacts with the alkyl iodide **1** to form radical *II* via XAT reaction. The radical is then involved in a 5-exo-trig and 5-exo-dig cascade cyclization to form the vinyl radical *IV* via tertiary radical *III*. Finally, a HAT reaction forms the hirsutene and regenerates the tin radical *I*.





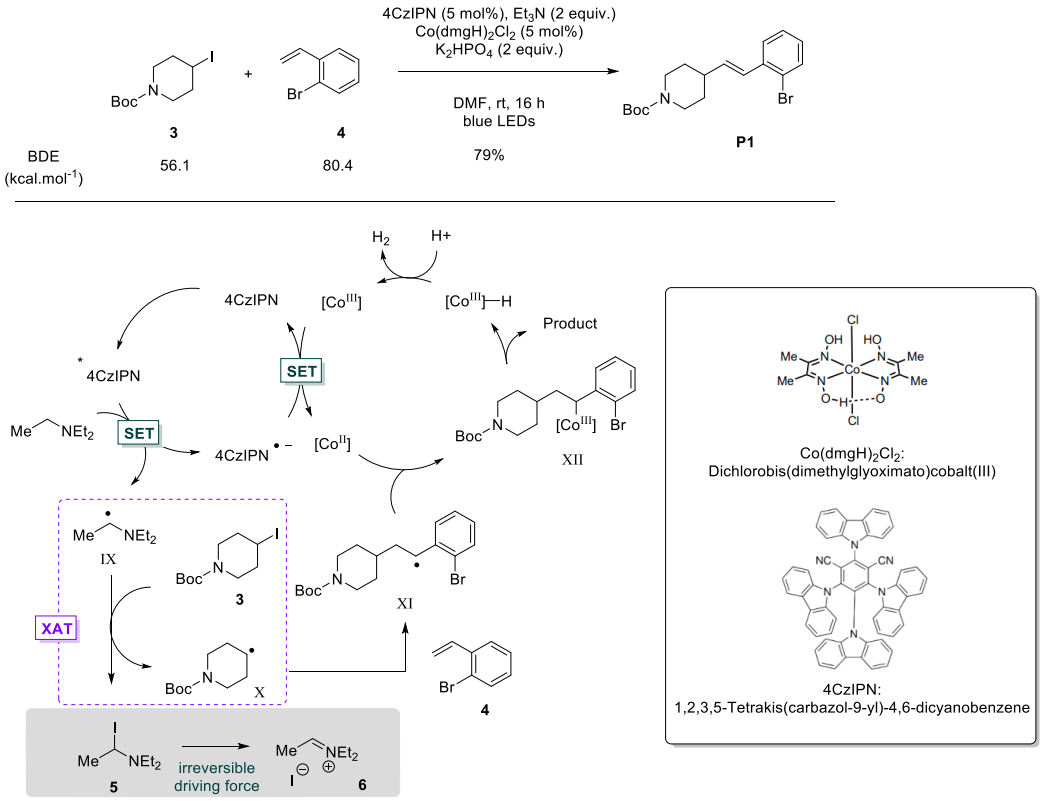
6. Viresh Rawal and Seiji Iwasa reported a photochemical rearrangement of α-iodo epoxide **2** (*Tetrahedron Lett.* **1992**, *33*, 4687-4690). Propose a mechanism.

Answer: Under high energy light irradiation (<400 nm), the tin dimer undergoes a Sn-Sn σ-bond homolysis to give the required tin radical *I* which reacts with the compound **2** to form radical *V* via XAT reaction. This radical undergoes a ring opening of the epoxide to form the oxo-radical *VI*. After 1,5-HAT reaction, the benzylic radical *VII* is involved in Atom Transfer Radical Cyclization (ATRC) to form radical *VIII* which is then involved in another XAT reaction with Bu_3SnI, generated earlier, to form the product and regenerate tin radical *I*.

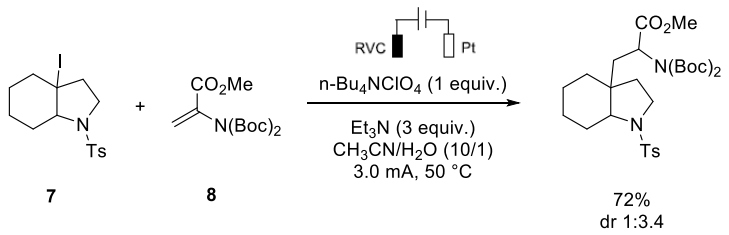


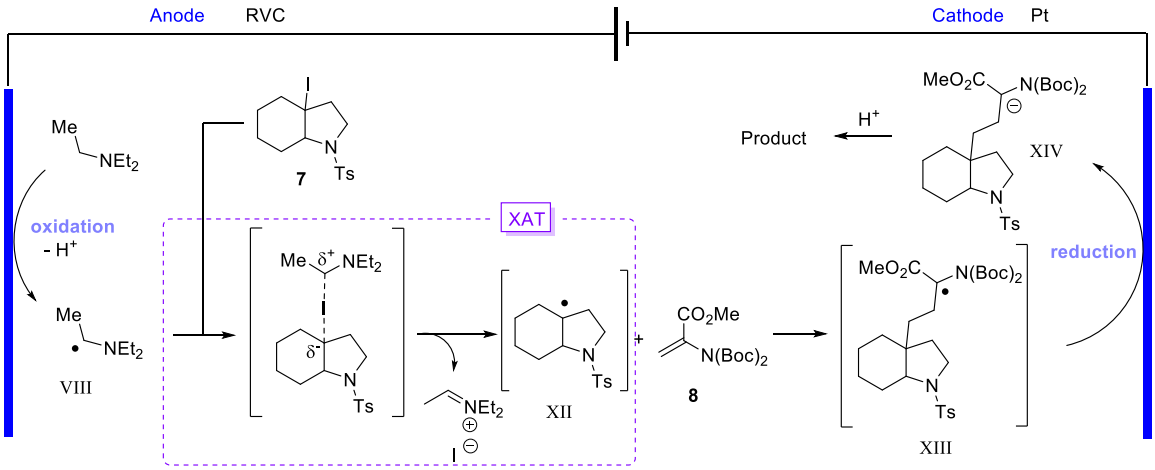
7. Leonori and Juliá developed an efficient strategy using nucleophilic α-aminoalkyl radicals and applied it in the following transformation (*Science*, **2020**, *367*, 1021-1026). Identify product **P1** and propose a mechanism of the reaction as well as an explanation of the selectivity towards alkyl iodide **3** over ArBr **4** based on the BDE provided.
<u>Answer</u>: Upon light irradiation, photocatalyst 4CzIPN gets excited into its triplet state and reacts with Et_3N in a SET reaction to form radical IX. *IX* is then involved in the XAT reaction with the alkyl iodide **3** to form radical *X* which adds on the double bond of the styrene **4** (Giese reaction) to form *XI*. $Co^{(II)}$, from reduction of the catalyst by the radical anion of the photocatalyst, forms the $Co^{(III)}$ species *XII* which gives the product and $Co^{(III)}$ hydride. $Co^{(III)}$ catalyst is regenerated by reacting with the proton source K_2HPO_4.
The XAT reaction occurs selectively with compound **3** over **4** due to its lower bond energy dissociation (BDE), 56.1 and 80.4 kcal/mol, respectively.

<u>Observation</u>: α-aminoalkyl radicals display a reactivity profile similar to that of tin radicals. The fast and irreversible dissociation of the α-iodoamine **5** into **6** provides the thermodynamic force to the process.



8. Xiang Sun and Ke Zheng have developed an electrochemical halogen-atom transfer (e-XAT) reaction (*Nature Commun.*, **2023**, *14*, 6825.) Propose a mechanism to form the radical.
   Answer: The α-aminoalkyl radical *IX* is formed via anodic oxidation followed by deprotonation. *IX* promotes then the homolytic activation of the C-I bond of **7** to form radical *XIII*. Next, the radical *XIII* adds to the electron-deficient alkene **8** to give the stabilised radical *XIV*. Finally, the product is formed after cathodic reduction of radical *XIV* into carbanion *XV* followed by protonation from water.


RVC
Pt
n-Bu4NClO4 (1 equiv.)
Et3N (3 equiv.)
CH3CN/H2O (10/1)
3.0 mA, 50 °C
CO2Me
N(Boc)2
Ts
7
8
72%
dr 1:3.4


Anode
RVC
Cathode
Pt
oxidation
- H+
reduction
XAT
Product
H+
VIII
XII
XIII
XIV
7
8